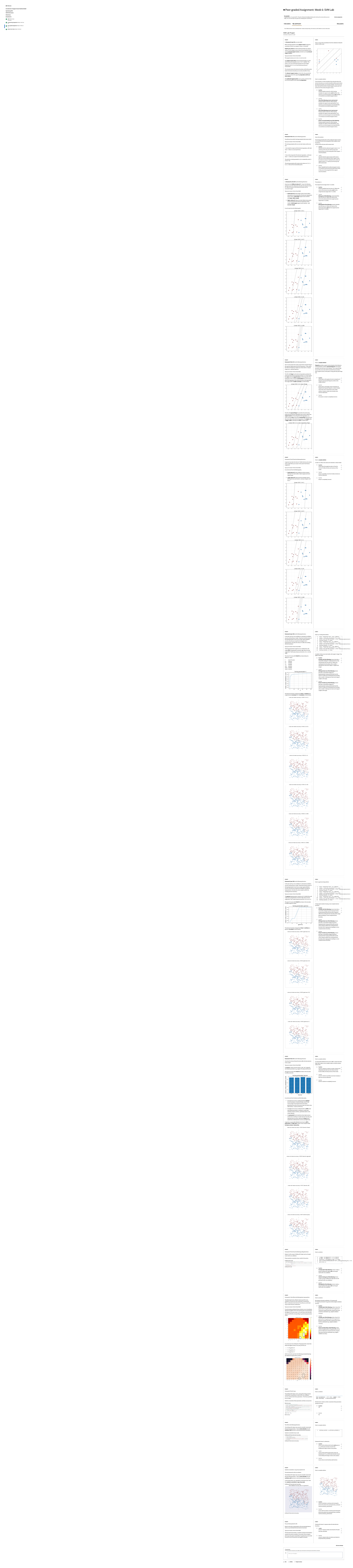

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Introduction to Support Vector Machine Model

Optimizing SVM

SVM Performance

Slide Decks

Assessments

- **Quiz:** Week 6 Quiz
Submitted
- **Programming Assignment:** Week 6: SVM Lab
3h
- **Peer-graded Assignment:** Week 6: SVM Lab
2h
- **Review Your Peers:** Week 6: SVM Lab

Peer-graded Assignment: Week 6: SVM Lab

You passed!
Congratulations. You earned 55 / 55 points. Review the feedback below and continue the course when you are ready. You can also help more peers by reviewing their submissions.

Review assignments

Instructions | My submission | Discussions

Your fellow learners have submitted their reviews anonymously. All names are still visible to course instructors.

SVM Lab Project

Submitted on November 18, 2023

PROMPT

In **Homework 6 part 1B**, you were asked:

Which training examples are the **support vectors**? Assign the coordinate in the list. e.g. support_vectors = [(1,0),(0.0)]

Explain your answer for which training examples are support vectors. For the support vectors that you have identified, which are the **in-bound support vectors** and which are the **out-bound support vectors**?

Type your answer in the rich text field.

The support vectors are [(-5, 0), (-5, 1), (6, 3), (5, 2)]

The **support vectors (SV)** are the training examples (xi, yi) for which we have the equation yi.(w.xi + b) <= 1 satisfied. The points (-5, 0), (5,2) are SVs because they lie on the margin hyperplane and (-5, 1), (6, 3) are SVs because they are misclassified.

The red points above the decision boundary and the blue points below the decision boundary are correctly classified. Hence,

The **inbound support vectors**: [(-5,0), (5,2)] , the ones that lie on the correct side of the separating **hyperplane (with zero slack values)**.

The **outbound support vectors**: [(-5, 1), (6, 3)] , the ones that lie on the wrong side of the separating **hyperplane**.

RUBRIC

Here is a copy of the visualization from the notebook (to help the solution make sense):

Here is a sample solution:

From the data it is a fair assumption that red points above the decision boundary are correctly classified and blue points below the decision boundary are correctly classified. From this we see that (-5,0) and (5,2) are the in-bound support vectors and (-5,1) and (6,3) are the out-bound support vectors.

- ● **3 points**
  Properly explains answer for which training examples are support vectors **and** identifies (-5,0) and (5,2) as in-bound support vectors **and** identifies (-5,1) and (6,3) as out-bound support vectors.
- ○ 2 points
  Two of the following correct, one incorrect: Properly explains answer for which training examples are support vectors **or** identifies (-5,0) and (5,2) as in-bound support vectors **or** identifies (-5,1) and (6,3) as out-bound support vectors.
- ○ 1 point
  One of the following correct, two incorrect: Properly explains answer for which training examples are support vectors **or** identifies (-5,0) and (5,2) as in-bound support vectors **or** identifies (-5,1) and (6,3) as out-bound support vectors.
- ○ 0 points
  Doesn't correctly identify any of the following: Properly explains answer for which training examples are support vectors **or** identifies (-5,0) and (5,2) as in-bound support vectors **or** identifies (-5,1) and (6,3) as out-bound support vectors.

PROMPT

Homework 6 Part 1C asks the following question:

How did you know which training examples had nonzero slack?

Type your answer in the rich text field.

The training examples with non-zero slack values are the ones that

1. lies inside the margin and the decision hyperplane, with slack value < 1 (there is no such point here)

OR

2. lies on the wrong side of the decision hyperplane., with slack value > = 1 (there are couple of misclassified points here)

The slack for a training example (xi, yi) is computed by max(0, 1 - yi*(w.xi + b)).

The training examples with nonzero slack values are: [(-5, 1), (6, 3)] - these are the misclassified points.

RUBRIC

Here is the solution:

The training examples that are the outbound support vectors, (-5,1) and (6,3) are on the wrong side of their support vector boundaries,
so they are the only ones with nonzero slack.

- ● **2 points**
  Identifies that the outbound support vectors (-5, 1) and (6,3) are the only ones with nonzero slack because they are on the wrong side of their support vector boundaries.
- ○ 1 point
  Identifies that the support vectors with nonzero slack are on the wrong side of their support vector boundaries but doesn't identify that the outbound support vectors (-5, 1) and (6,3) are the only ones with nonzero slack.
- ○ 0 points
  Doesn't identify that the outbound support vectors (-5,1) and (6,3) are the only ones with nonzero slack or they are on the wrong side of their support vector boundaries.

PROMPT

In **Homework 6, Part 2B** has the following directions:

Experiment with **different values of C**. Answer the following question in your Peer Review for the week: How does the choice of C affect the nature of the decision boundary and the associated margin?

Type your answer in the rich text field.

1. **Small values of C** allows higher values of slack (allows more points on the wrong side of the decision hyperplane) which creates a **soft-margin** support vector machine - with **larger (wider) margin**
2. **Higher values of C** allows less slack (allows lesser points on the wrong side of the decision hyperplane) which creates a **hard-margin** support vector machine - with **narrower margin**

It can be seen from the following plot:

Linear SVC C=0.1

Linear SVC C=0.5

Linear SVC C=1

Linear SVC C=10

Linear SVC C=100

RUBRIC

The solution is:

The margin becomes larger when C is smaller.

- ● **5 points**
  Properly identifies how the choice of C affects the nature of the decision boundary **and** that the margin becomes larger when C is smaller.
- ○ 3 points
  Missing one of the following: properly identifies how the choice of C affects the nature of the decision boundary **or** that the margin becomes larger when C is smaller.
- ○ 0 points
  Missing both of the following: properly identifies how the choice of C affects the nature of the decision boundary **and** that the margin becomes larger when C is smaller.

PROMPT

Homework 6 Part 2C has the following directions:

Set C=3 and compare the results you get when using the hinge vs the squared_hinge values for the loss parameter. In this week's Peer Review Assignment: Explain your observations. Compare hinge loss vs. squared hinge loss.

Type your answer in the rich text field.

The SVC with **hinge** loss the decision hyperplane penalizes the points from the decision hyperplane linearly, as a result it allows less **slack** and more **support vectors** on the wrong side of the hyperplane, since a red point **misclassified** (lies on the wrong side of the hyperplane) and a blue point lies on the hyperplane. The margin width is **smaller** (**narrower**), as shown below.

Linear SVC C=3, loss=hinge

The SVC with **squared hinge** loss penalizes the points lying further from the decision hyperplane more, hence it allows less **support vectors** on the wrong side of the hyperplane, as a result, we have **none** of the points **misclassified** (all points lie on the correct sides of the decision hyperplane). The **margin** width is **bigger** (**wider**), allowing more **slack**, as shown below.

Linear SVC C=3, loss=squared_hinge

RUBRIC

Here is a **sample solution**:

Hinge loss penalizes support vectors linearly to their distances to the hyperplane, whereas **squared hinge loss** more heavily penalizes ones that have more violations. Thus, squared hinge loss provides more slack within the dashed lines but allows fewer support vectors on the plane's wrong side than plain hinge loss.

- ● **5 points**
  Answer gives a thorough and correct comparison of hinge loss and squared hinge loss similar to the sample solution.
- ○ 3 points
  Answer gives a somewhat correct comparison of hinge loss and squared hinge loss- answer includes some of the same components as the sample solution. However, answer also includes some incorrect statements.
- ○ 0 points
  No answer or answer is completely incorrect.

PROMPT

Homework 6 Part 2D has the following directions:

In general, how does the choice of C affect the bias and variance of the model? Write your answer in this week's Peer Review assignment.

Type your answer in the rich text field.

As can be seen from the following plots,

1. **Small values of C** lead to high bias and low variance, allowing more slack (softer / wider margins) and more robust model
2. **Larger values of C** lead to low bias and high variance, allowing for less slack (harder /narrower margins) and less ro

Linear SVC C=0.1

Linear SVC C=0.5

Linear SVC C=1

Linear SVC C=10

Linear SVC C=100

RUBRIC

Here is a **sample solution**:

A small c can allow more slack and is therefore a robust model.

- ● **5 points**
  Answer gives thorough description of how the choice of C affects the bias and variance of the model.
- ○ 3 points
  Answer is partially correct but includes at least one incorrect statement.
- ○ 0 points
  Answer is completely incorrect.

PROMPT

Homework 6 part 3B has the following directions:

In this part we'll use cross-validation to estimate the validation accuracy achieved by our model. Experiment with the values of the hyperparameters to see if you can get a good validation accuracy. Describe your experiment strategy for the C hyperparamter. How does the choice of C affect the resulting decision boundary?

Type your answer in the rich text field.

The **C** hyperparameter is kept low at 0.1 initially (from soft margin **SVC**) and gradually increased to high values (to hard margin SVC). The **C** values chosen are from [0.1, 0.5, 1, 10, 100, 1000].

The mean CV scores with **5-fold CV** are shown below, for different C values:

```
C       mean CV scores
0.1     0.903333
0.5     0.910000
1.0     0.920000
10.0    0.933333
100.0   0.940000
1000.0  0.943333
```

Tuning paramater C

The decision boundary changes from **linear** to **nonlinear** (the margin becomes **narrower**) as C is **increased**, as shown below.

cross-val mean-accuracy: 0.903 (C=0.1)

cross-val mean-accuracy: 0.910 (C=0.5)

cross-val mean-accuracy: 0.920 (C=1)

cross-val mean-accuracy: 0.933 (C=10)

cross-val mean-accuracy: 0.940 (C=100)

cross-val mean-accuracy: 0.943 (C=1000)

RUBRIC

Here is a C tuning code solution:

```
nlsvm = SVC(kernel="rbf", C=0.1, gamma=1)
scores = cross_val_score(nlsvm, X, y, cv=5)
print("cross-val mean-accuracy: {:.3f}".format(np.mean(scores)))
nonlinear_plot(X, y, nlsvm)
nlsvm = SVC(kernel="rbf", C=1, gamma=1)
scores = cross_val_score(nlsvm, X, y, cv=5)
print("cross-val mean-accuracy: {:.3f}".format(np.mean(scores)))
nonlinear_plot(X, y, nlsvm)
nlsvm = SVC(kernel="rbf", C=10, gamma=1)
scores = cross_val_score(nlsvm, X, y, cv=5)
print("cross-val mean-accuracy: {:.3f}".format(np.mean(scores)))
nonlinear_plot(X, y, nlsvm)
```

A smaller C allows more slack (wider soft margin). A larger C has tighter soft margin.

- ● **5 points**
  Includes all of the following: learner describes a reasonable strategy for C hyperparameter tuning and describes how the choice of C affects the resulting decision boundary with a smaller C allowing more slack and a larger C, a tighter soft margin.
- ○ 3 points
  Missing at least one of the following: learner describes a reasonable strategy for C hyperparameter tuning and describes how the choice of C affects the resulting decision boundary with a smaller C allowing more slack and a larger C, a tighter soft margin.
- ○ 0 points
  Doesn't include any of the following: learner describes a reasonable strategy for C hyperparameter tuning and describes how the choice of C affects the resulting decision boundary with a smaller C allowing more slack and a larger C, a tighter soft margin.

PROMPT

Homework 6 part 3B has the following directions:

In this part we'll use cross-validation to estimate the validation accuracy achieved by our model. Experiment with the values of the hyperparameters to see if you can get a good validation accuracy. Describe your experiment strategy for the gamma hyperparamter. How does the choice of gamma affect the resulting decision boundary?

Type your answer in the rich text field.

The **gamma** hyperparameter is kept low at 0.1 initially (from soft margin SVC) and gradually increased to high values (to hard margin SVC). The C values chosen are from [0.1, 0.2, 0.3, 0.5, 1].

The mean CV scores with **5-fold CV** are shown in the next plot, for different gamma values:

Tuning paramater gamma

The decision boundary changes from **linear** to **nonlinear** as gamma is **increased**, as shown below.

cross-val mean-accuracy: 0.907 (gamma=0.1)

cross-val mean-accuracy: 0.910 (gamma=0.2)

cross-val mean-accuracy: 0.920 (gamma=0.5)

cross-val mean-accuracy: 0.920 (gamma=1)

RUBRIC

Here is a gamma tuning solution:

```
nlsvm = SVC(kernel="rbf", C=1, gamma=5)
scores = cross_val_score(nlsvm, X, y, cv=5)
print("cross-val mean-accuracy: {:.3f}".format(np.mean(scores)))
nonlinear_plot(X, y, nlsvm)
nlsvm = SVC(kernel="rbf", C=1, gamma=10)
scores = cross_val_score(nlsvm, X, y, cv=5)
print("cross-val mean-accuracy: {:.3f}".format(np.mean(scores)))
nonlinear_plot(X, y, nlsvm)
nlsvm = SVC(kernel="rbf", C=1, gamma=50)
scores = cross_val_score(nlsvm, X, y, cv=5)
print("cross-val mean-accuracy: {:.3f}".format(np.mean(scores)))
nonlinear_plot(X, y, nlsvm)
```

A larger gamma leads to having a more complex decision boundary

- ● **5 points**
  Includes all of the following: learner describes a reasonable strategy for gamma hyperparameter tuning and describes how the choice of gamma affects the resulting decision boundary with a larger gamma leading to a more complex decision boundary.
- ○ 3 points
  Missing at least one of the following: learner describes a reasonable strategy for gamma hyperparameter tuning and describes how the choice of gamma affects the resulting decision boundary with a larger gamma leading to a more complex decision boundary.
- ○ 0 points
  Doesn't include any of the following: learner describes a reasonable strategy for gamma hyperparameter tuning and describes how the choice of gamma affects the resulting decision boundary with a larger gamma leading to a more complex decision boundary.

PROMPT

Homework 6 Part 3C has the following directions:

How does the choice of kernel function affect the bias/variance of the model?

Type your answer in the rich text field.

The **kernel** is chosen are from ['linear', 'poly', 'rbf', 'sigmoid'], with default parameters (e.g., degree 3 for polynomial kernel).

The mean CV scores with **5-fold CV** are shown in the next plot, for different kernels:

Tuning paramater kernel

As can be seen from the above and the below plots,

- the lowest CV accuracy is obtained with the **sigmoid** kernel (resulting in a model with high variance, with narrow margin and complex decision boundary- generalizes less well on the unseen data, the model is also with low bias - it seems to fit the tra).
- the highest CV accuracy is obtained with .the **RBF** kernel (with default parameters, resulting in a model with moderate variance and bias - generalizes better on the unseen dataset),.
- the **polynomial** kernel model also shows high variance and low bias (resulting in complex decision boundary and relatively lower accuracy), whereas the **linear** kernel model shows relatively lower variance and higher bias.

In general, an expressible (flexible) kernel (such as **RBF** or **polynomial** with **high order**) makes a more complex model- **increases variance**, **reduces bias**.

cross-val mean-accuracy: 0.910 (kernel=linear)

cross-val mean-accuracy: 0.903 (kernel=sigmoid)

cross-val mean-accuracy: 0.937 (kernel=rbf)

cross-val mean-accuracy: 0.907 (kernel=poly)

RUBRIC

Here is a sample solution:

An expressible (flexible) kernel (such as RBF or polynomial with high order) makes a more complex model- increases variance, reduces bias.

- ● **5 points**
  Learner's solution is similar to sample. Properly and adequately describes how the choice of kernel function affects the bias/variance of the model.
- ○ 3 points
  Learner's solution is partially correct but includes at least one incorrect statement.
- ○ 0 points
  Learner's solution is completely incorrect.

PROMPT

Homework 6 Part 4A has the following coding directions:

Replace it with a proper GridSearchCV object and run the grid search with cross validation.

Please upload a screenshot of your code for this section.

GridSearchCV code

GridSearchCV code

RUBRIC

Here is a solution:

```
c_range = np.logspace(-5, 5, 11, base=2)
g_range = np.logspace(-5, 5, 11, base=2)
param_grid = dict(gamma=g_range, C=c_range)
grid = GridSearchCV(SVC(kernel="rbf"), param_grid=param_grid, cv=3)
grid.fit(X, y)
```

- ● **5 points**
  Includes both of the following: Learner creates a proper GridSearchCV object **and** runs the grid search with cross validation.
- ○ 3 points
  Learner missing one of the following: learner creates a proper GridSearchCV object **or** runs the grid search with cross validation.
- ○ 0 points
  Missing both of the following: Learner creates a proper GridSearchCV object **and** runs the grid search with cross validation.

PROMPT

Homework 6, Part 4B has the following heat-map question:

The following function will plot a heat-map of the cross-validation accuracies for each combination of parameters. Which combination looks the best? How did you use the heat-map to select the best combination?

Type your answer in the rich text field.

From the heatmap obtained (shown below), we can immediately identify the brightest cells in the heatmap plot (corresponding to the highest CV accuracy value) - let's mark those cells with black squares and find the corresponding (C, gamma) hyperparameter values.

Validation accuracy

As can be seen, the combination of hyperparameter values that obtain the highest (mean) CV accuracy (94.3%) are:

- C = 32, gamma = 2
- C = 16, gamma = 8
- C = 8, gamma = 8

which can also be seen from the following annotated heatmap plot obtained using the library seaborn:

Validation Accuracy

RUBRIC

Here is a solution:

By looking at the grid visualization, (C=8, gamma=8), (C=16,gamma=8) and (C=32, gamma=2) have highest validation accuracy.

- ● **5 points**
  Includes both of the following: Author chose (C=8, gamma=8), (C=16, gamma=8) or (C=32, gamma=2) and learner properly describes using the heat-map to find the combination (e.g., highest validation accuracy).
- ○ 3 points
  Includes one of the following: Author chose (C=8, gamma=8), (C=16, gamma=8) or (C=32, gamma=2) **or** learner properly describes using the heat-map to find the combination (e.g., highest validation accuracy).
- ○ 0 points
  Doesn't include either of the following: Author chose (C=8, gamma=8), (C=16, gamma=8) or (C=32, gamma=2) **or** learner properly describes using the heat-map to find the combination (e.g., highest validation accuracy).

PROMPT

Homework 6 Part 4C says:

The GridSearchCV object scores, among other things, the best combination of parameters as well as the cross-validation accuracy achieved with those parameters. Print those quantities for our model.

Submit a screenshot of best parameters and best accuracy here.

Best Accuracy

Best Accuracy

RUBRIC

Here is a solution:

```
best parameters:  {'C': 8.0, 'gamma': 8.0}
best accuracy:  0.9433333333333334
```

Did the learner submit a similar screenshot of best parameters and best accuracy?

- ● **5 points**
  Yes
- ○ 0 points
  No

PROMPT

Part 4D has the following directions:

The GridSearchCV object also stores the classifier trained with the best hyperparameters. Pass this **best estimator** into the **nonlinear_plot** function to view the best decision boundary.

Upload a screenshot of your code.

GridSearchCV best decision boundary

GridSearchCV best decision boundary

RUBRIC

Here is a sample solution:

```
nonlinear_plot(X, y, grid.best_estimator_)
```

Evaluate the learner's submission.

- ● **2 points**
  Learner calls nonlinear plot function **and** passes in X, y, and the best estimator stored in the GridSearchCV object similar to the solution.
- ○ 1 point
  Learner calls nonlinear plot function. Does not properly pass in X, y, and the best estimator stored in the GridSearchCV object similar to the solution
- ○ 0 points
  Learner does not call nonlinear plot function.

PROMPT

Upload a screenshot or copy of your plot for 4D.

The lab directions for 4D are as follows:

The GridSearchCV object also stores the classifier trained with the best hyperparameters. Pass this **best estimator** into the **nonlinear_plot** function to view the best decision boundary.

In the prompt above, you uploaded a screenshot of your code. Now **upload a screenshot or copy of your plot.**

GridSearchCV best decision boundary

Best Validation Accuracy with C=8, gamma=8

GridSearchCV best decision boundary

RUBRIC

Here is a sample solution:

- ● **2 points**
  Learner submitted a nonlinear plot showing the decision boundary from passing the best_estimator into the nonlinear_plot function.
- ○ 0 points
  Learner did not submit a nonlinear plot showing the decision boundary from passing the best_estimator into the nonlinear_plot function.

PROMPT

For your final question for 4D:

Based on the plot you generated in 4D (and submitted above), what can you tell about the best decision boundary?

Type your answer in the rich text field.

The best decision boundary is nonlinear and goes horizontally to separate the positive examples, it has pretty narrow margins, with only a few points misclassified. It separates the convex hulls formed of most of the positive and negative examples.

RUBRIC

Evaluate the learner's response about the best decision boundary.

- ● **1 point**
  Learner's response makes sense based on the plot that they submitted.
- ○ 0 points
  Learner's response does not make sense based on the plot that they submitted.

Start new attempt

Comments

Comments left for the learner are visible only to that learner and the person who left the comment.

SD

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